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# SC2107/CE2107
# Microprocessor System Design and Development

## Serial Bus Interface

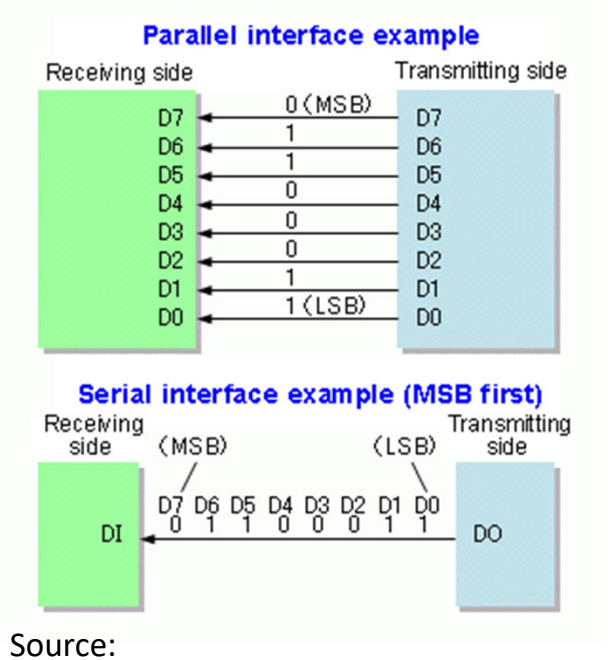
# Topics Covered

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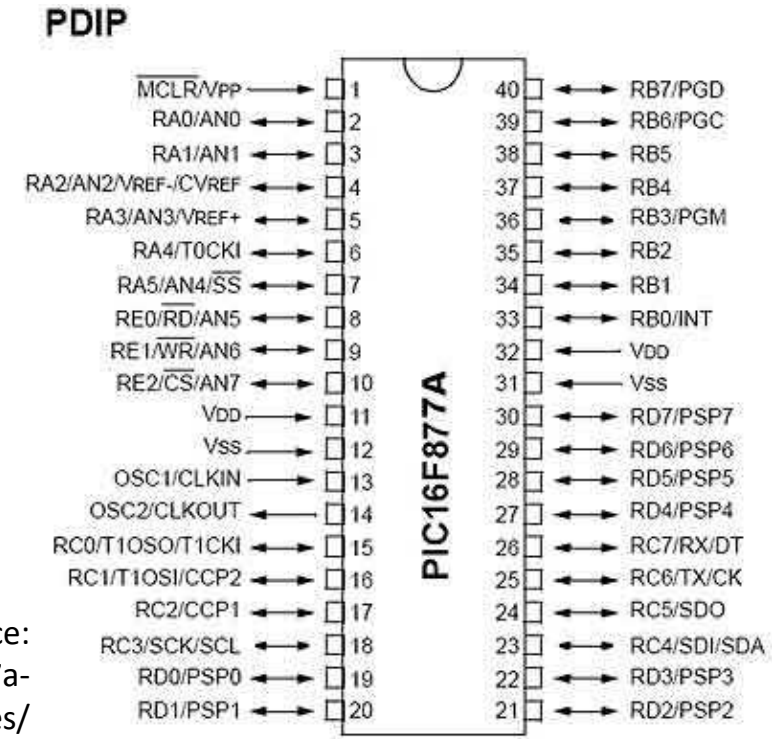
- Serial Communication
  - UART
  - SPI
  - I2C
  - I2S
  - NEC IR
  - USB

# Serial Communication

- Serial Communication are very common in embedded systems as they help to reduce the number of wires needed for interfacing.
- Common serial interfaces include:
  - UART
  - I2C
  - SPI
  - NEC IR
- Many peripheral devices usually have one or more of these interface options.




Source: https://en.wikipedia.org/wiki/Parallel_communication

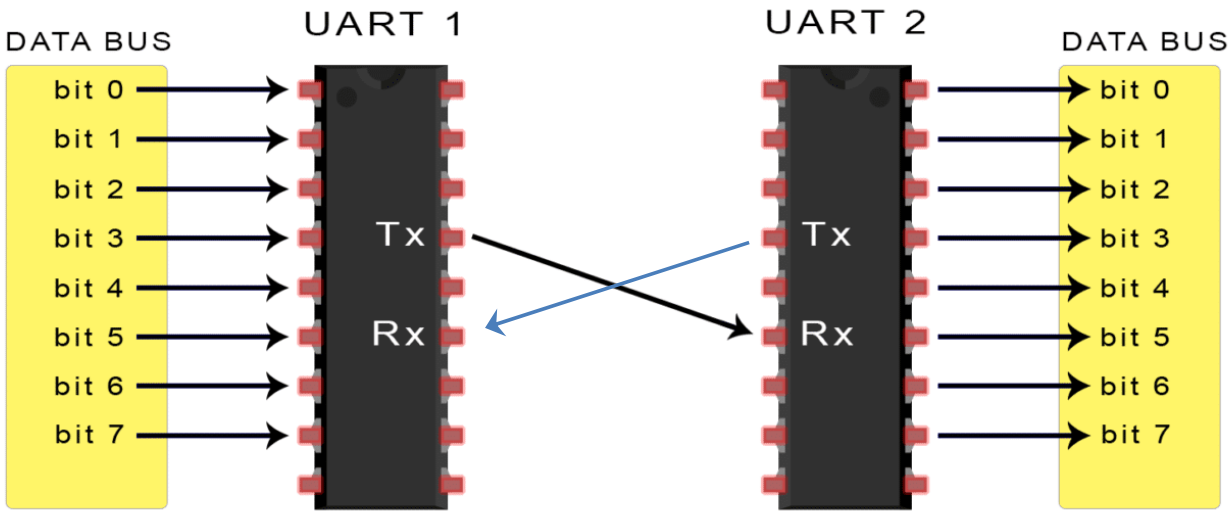



Source: http://microcontrollerslab.com/pic16f877a-introduction-features/

# Universal Asynchronous Receiver/Transmitter (UART)

# Universal Asynchronous Receiver/Transmitter

- UART is physical circuit in a microcontroller, or a stand-alone IC for transmitting and receiving serial data.

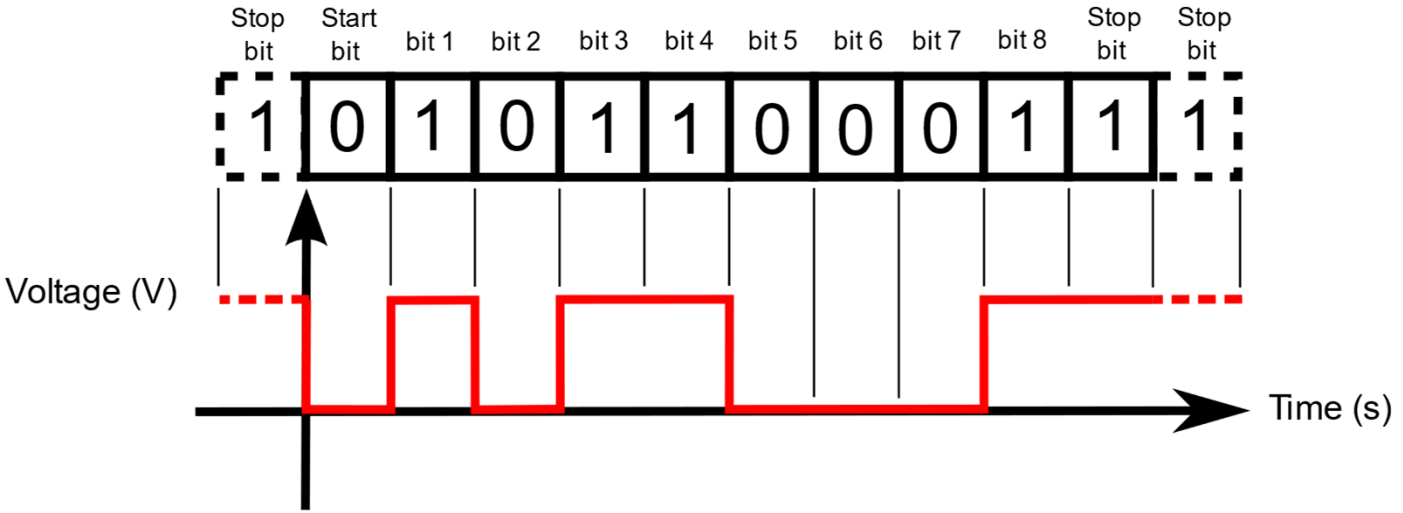



Source: http://www.circuitbasics.com/basics-uart-communication/

- Only two wires required to send data between two UARTs asynchronously (no clock).
- UARTs must agree on a bit rate (9600, 19200, 38400, -57600 or 115200 bits/sec) to communicate. The bitrate configurability makes it universal

# UART Byte Framing

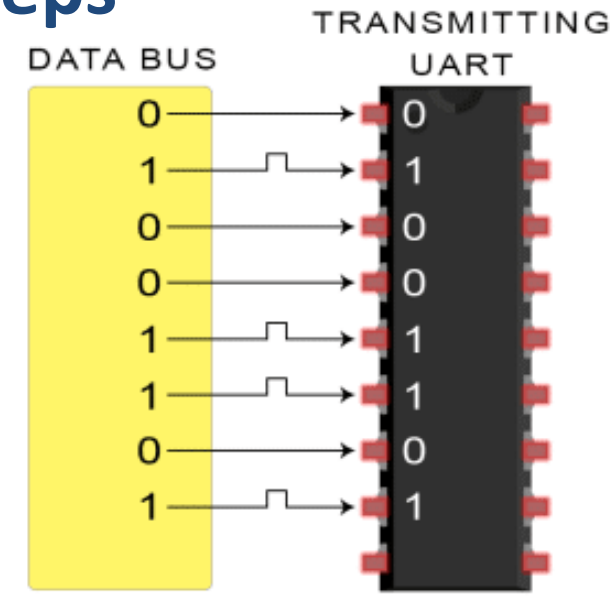
- Data is transferred in the following way:
  - In idle state, data line has high logic level ‘1’.
  - Each data transmission starts with a START bit, which is always a ‘0’.
  - Each data is 8- or 9-bit wide (LSB is transferred first)
  - Optional parity bit.
  - Each data transmission ends with a STOP bit, which is always a ‘1’.

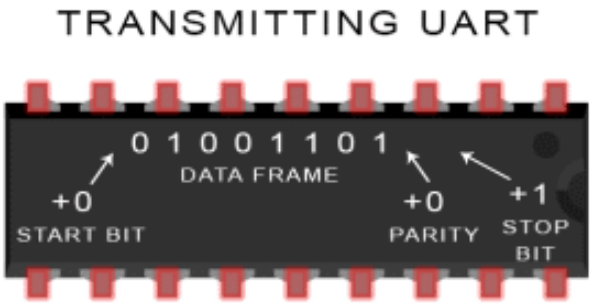


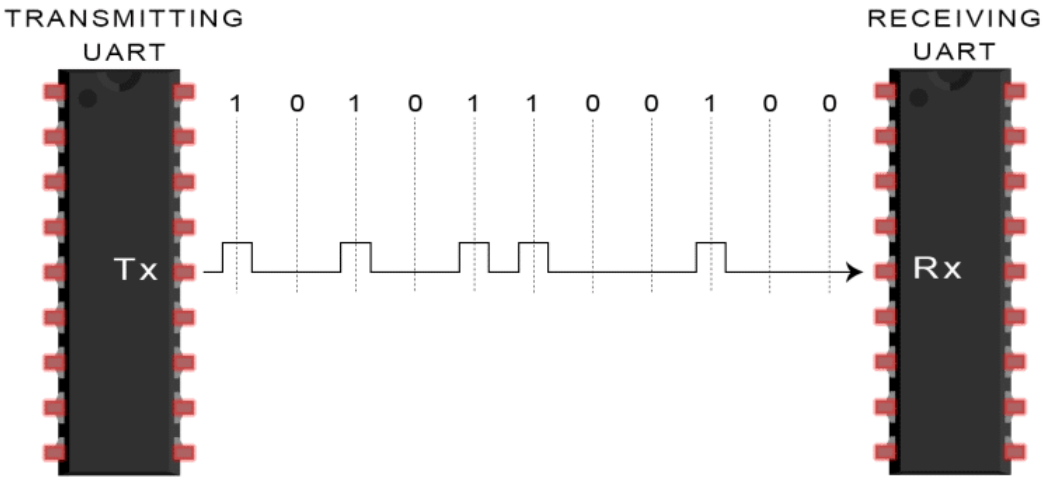

Source: http://www.circuitgrove.com/tutorials/uart-universal-asynchronous-receivertransmitter

# UART Transmission Steps

1. Transmitting UART receives data in parallel from the data bus.
2. Transmitting UART adds the start bit, parity bit, and the stop bit(s) to the data frame.
3. Entire packet is sent serially from the transmitting UART to the receiving UART. The receiving UART samples the data line at the pre-configured baud rate.



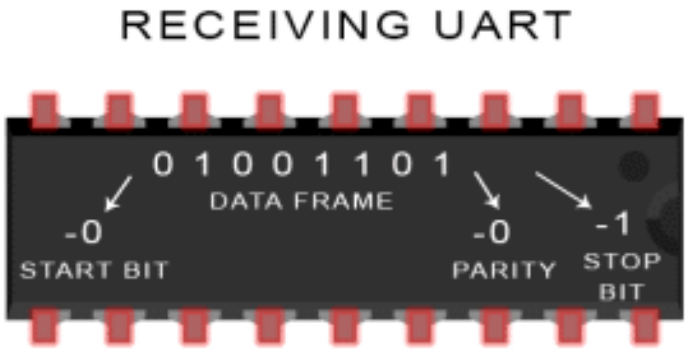



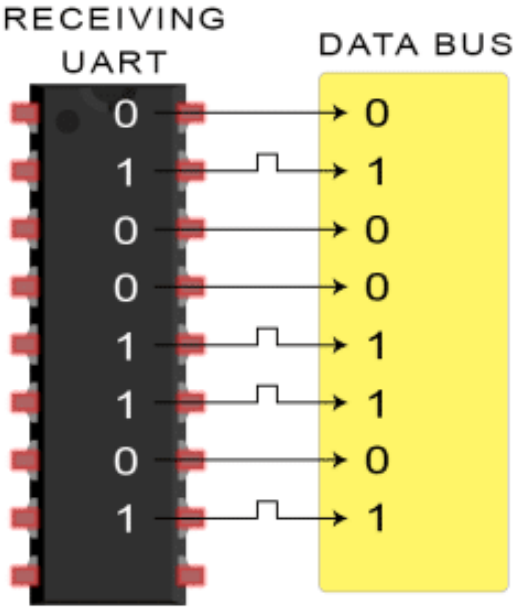




Source: http://www.circuitbasics.com/basics-uart-communication/

# UART Transmission Steps (Cont.)

4. Receiving UART discards the start bit, parity bit, and stop bit from the data frame.
5. Receiving UART converts the serial data back into parallel and transfers it to the data bus on the receiving end.

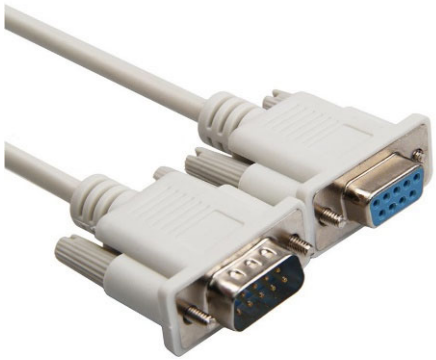


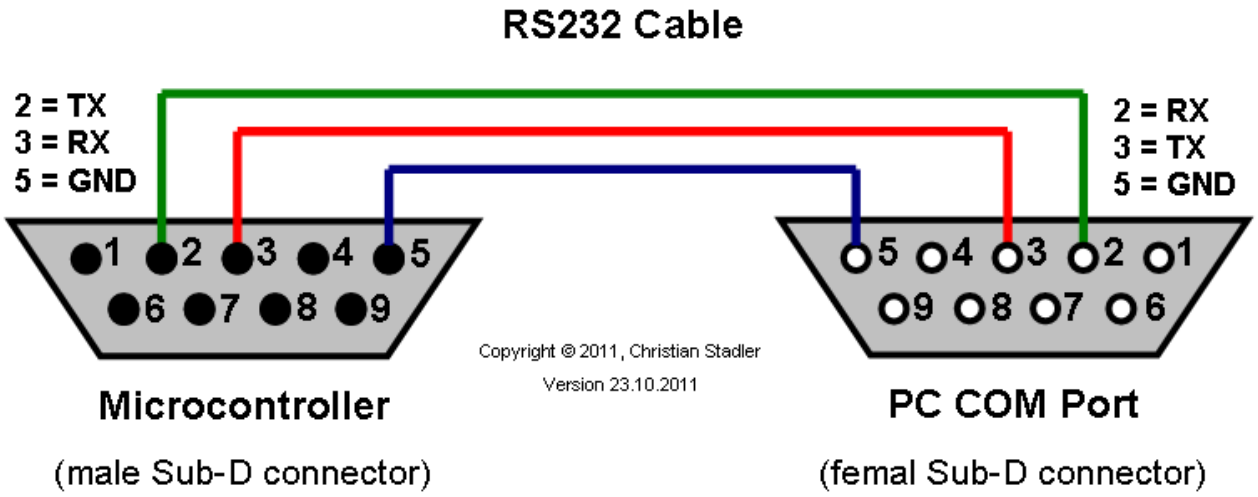




Source: http://www.circuitbasics.com/basics-uart-communication/

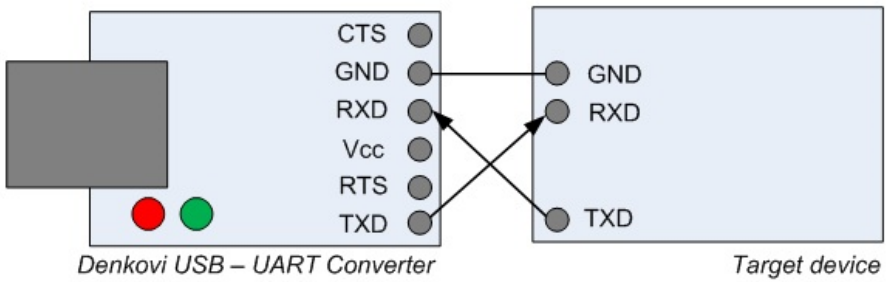
# RS232 Cable and FTDI Chip

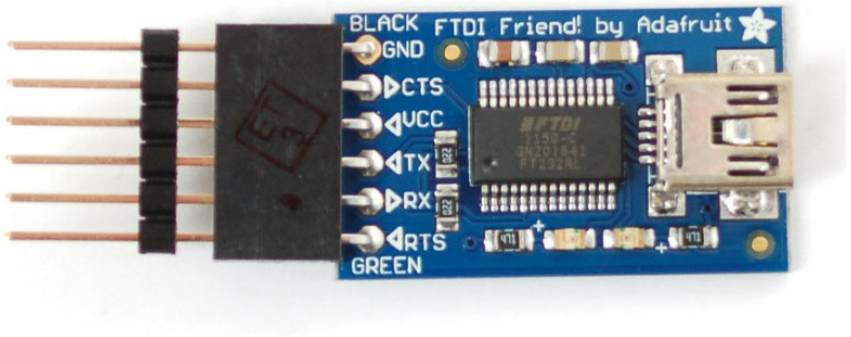
- Serial port (not common now).






- FTDI Chip.



# Advantages/Disadvantages of UART

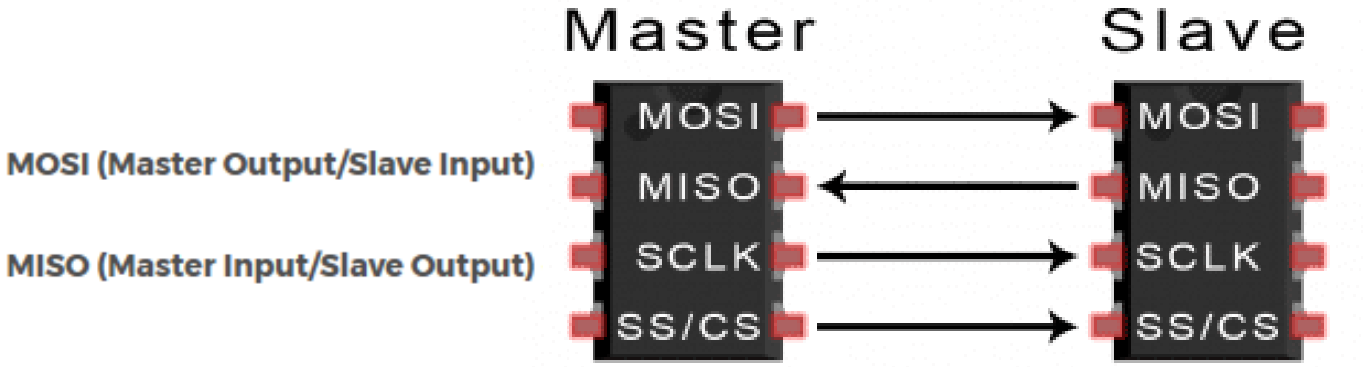
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- Advantages:
  - Only uses two wires
  - No clock signal is necessary
  - Has a parity bit to allow for error checking
  - The structure of the data packet can be changed
  - Well documented and widely used method
- Disadvantages:
  - The size of the data frame is limited to a maximum of 9 bits
  - Does not support multiple slave or multiple master systems

# Serial Peripheral Interface (SPI)

# Serial Peripheral Interface (SPI)

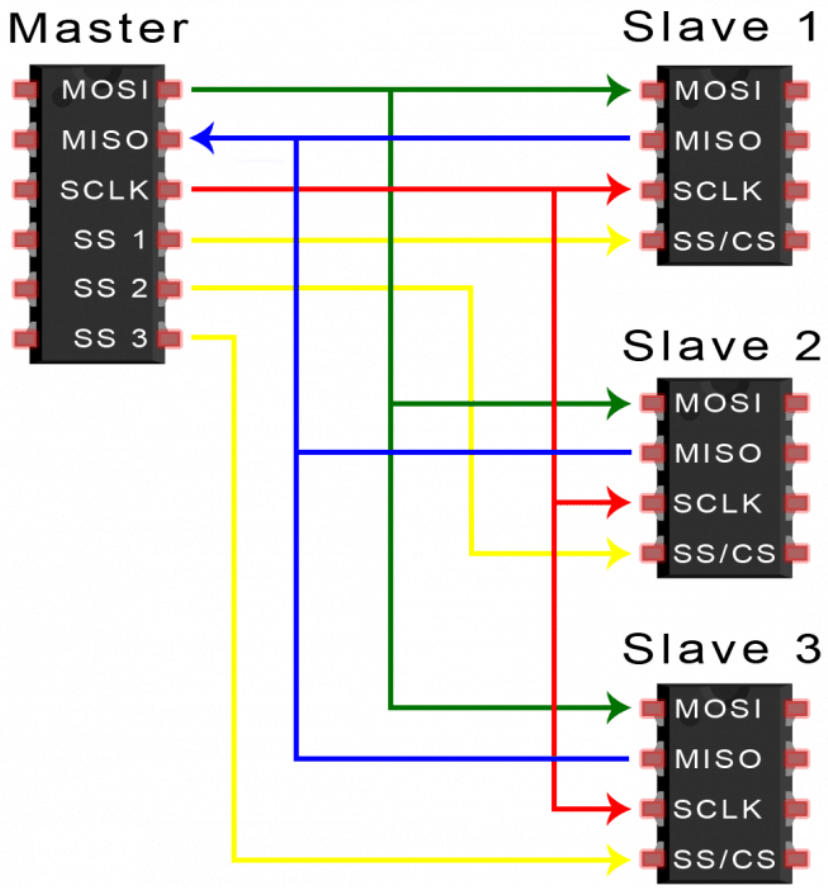
- Common communication protocol to communicate with microcontrollers, for example, SD card modules, RFID card reader modules, and 2.4 GHz wireless transmitter/receivers.
- Any number of bits can be sent or received in a continuous stream (up to 10Mbps).
- Master-slave communication. Only one master allowed. One master can control more than one slave.




Source: http://www.circuitbasics.com/basics-of-the-spi-communication-protocol

# Connecting to Multiple Slaves

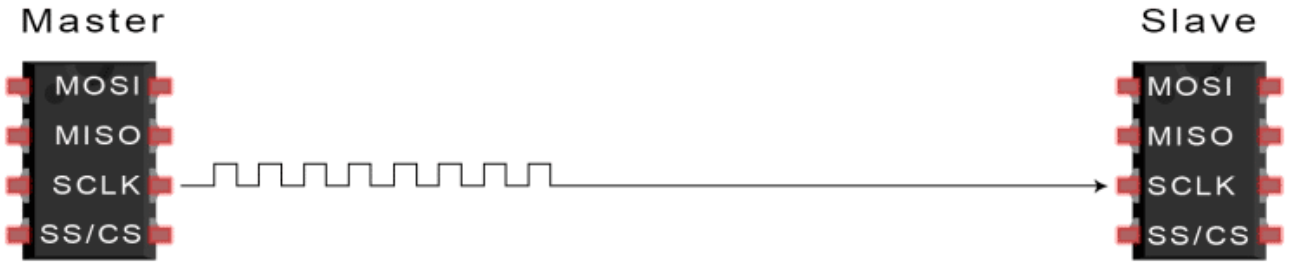
- SPI communication is always initiated by the master.
- Master generates the clock signal and one bit of data is transferred in each clock cycle.
- Master chooses slave it wants to talk to by setting the slave's CS/SS line to LOW.
- Master sends data to the slave serially through the MOSI line.
- Slave sends data serially back to the master through the MISO line.



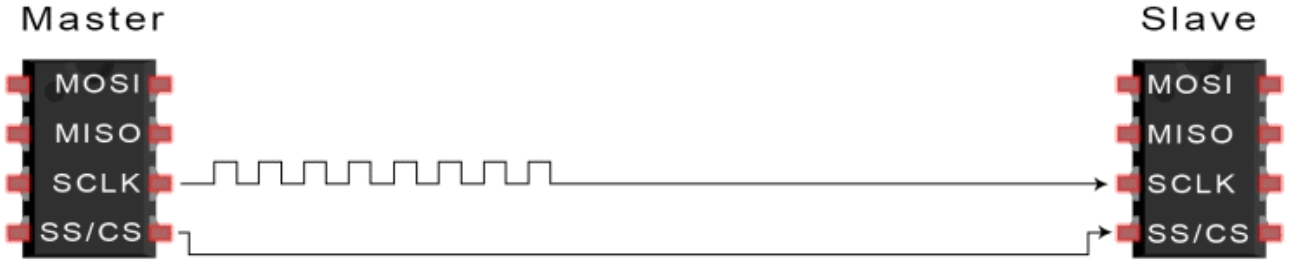

Source: http://www.circuitbasics.com/basics-of-the-spi-communication-protocol

# SPI Data Transmission Steps

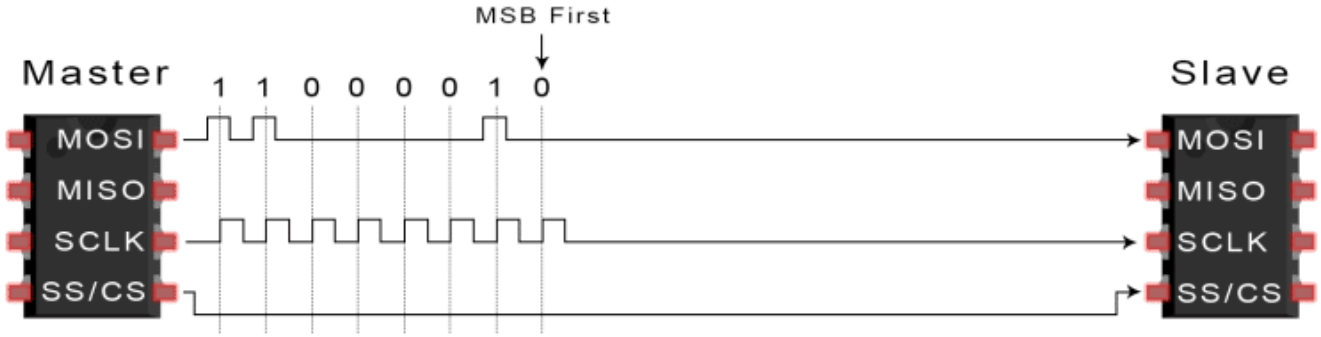
1. Master outputs the clock signal.




2. Master switches the SS/CS pin to LOW to activate the slave.




3. Master sends the data one bit at a time to the slave along the MOSI line.



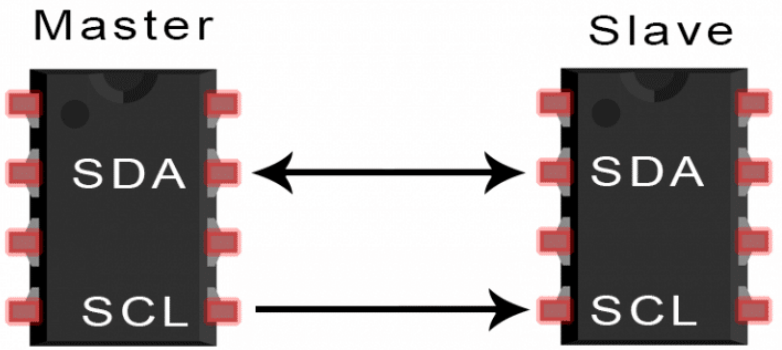
# Advantages/Disadvantages of SPI

- Advantages:
  - No start and stop bits, so the data can be streamed continuously without interruption.
  - No complicated slave addressing system.
  - Higher data transfer rate than UART and I2C.
  - Separate MISO and MOSI lines, so data can be sent and received at the same time
- Disadvantages:
  - Uses four wires (I2C and UARTs use two).
  - No acknowledgement that the data has been successfully received.
  - No form of error checking like the parity bit in UART.
  - Only allows for a single master.

# Inter-Integrated Circuit (I2C)

# Inter-Integrated Circuit (I2C)

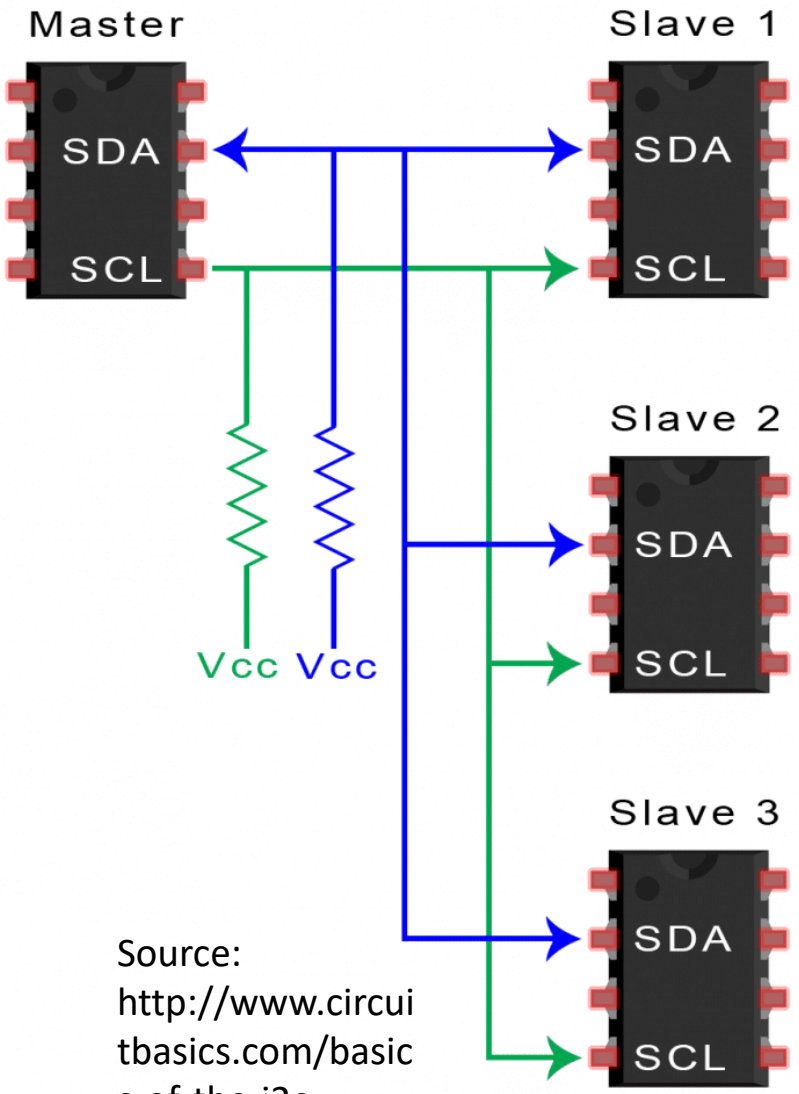
- Commonly used for communicating with on-board devices, e.g. OLED displays, barometric pressure sensors, or gyroscope/accelerometer modules.
- Data is transferred in messages (up to 400Kbps).
- Master-slave communication. Multiple masters are allowed. Each master can control more than one slave.
- Only two wires are used for data transfer:
  - SDA: Serial Data
  - SCL: Serial Clock




Source: http://www.circuitbasics.com/basics-of-the-i2c-communication-protocol/

# Connecting to Multiple Slaves

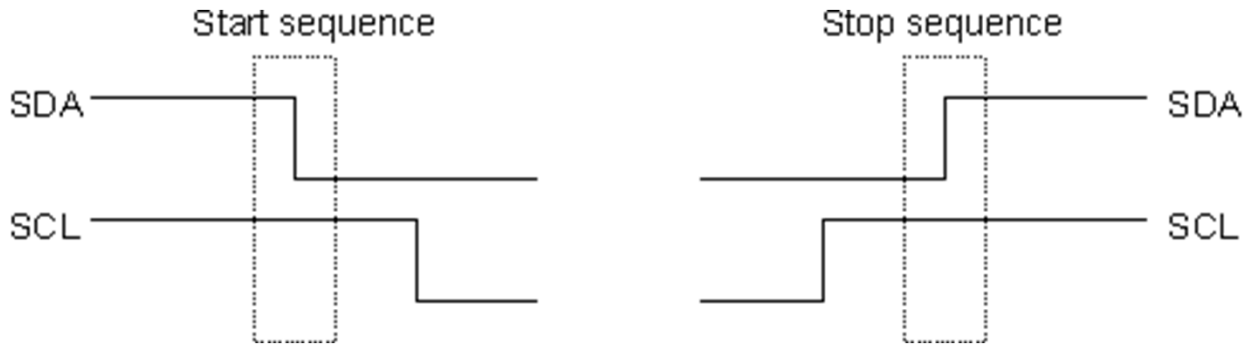
- I2C communication is initiated by the master.
- Master generates the clock signal (SCL) and one bit of data is transferred in each clock cycle.
- Master chooses slave it wants to talk to, using addressing (usually 7-bit address).
- Master and slave sends data through the SDA line.



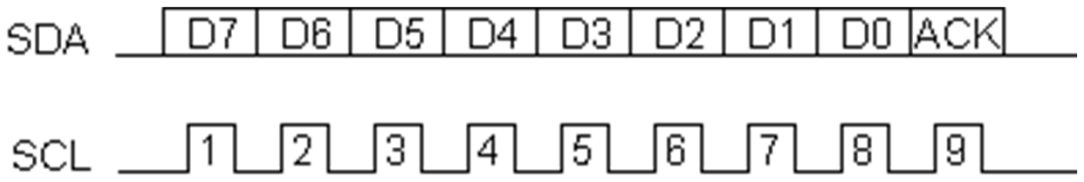

Source: http://www.circuitbasics.com/basics-of-the-i2c-communication-protocol/

# I2C Physical Protocol

- When the Master wishes to talk to a Slave, it initiates a ‘Start’ sequence on the bus.
- The ‘Start’ and ‘Stop’ are two special sequences on the I2C bus where the SDA line is allowed to change when the SCL is high.




- When data is being transferred, the SDA line must remain stable and not change whilst SCL is high.



# I2C Device Addressing

- All I2C devices are address using 7-bits or 10-bits. 7-bit addresses are the most common.
- When sending out the address, we send out 8-bits.
- 8th bit is used to indicate Read/Write operation.
- The 7-bit address is placed in the upper 7-bits of the byte and the Read/Write ($R/\overline{W}$) bit is in the LSB.




- The 7-bit addressing can cause some confusion so it is best to think of it as an 8-bit address.
  - Odd address as Read
  - Even address as Write

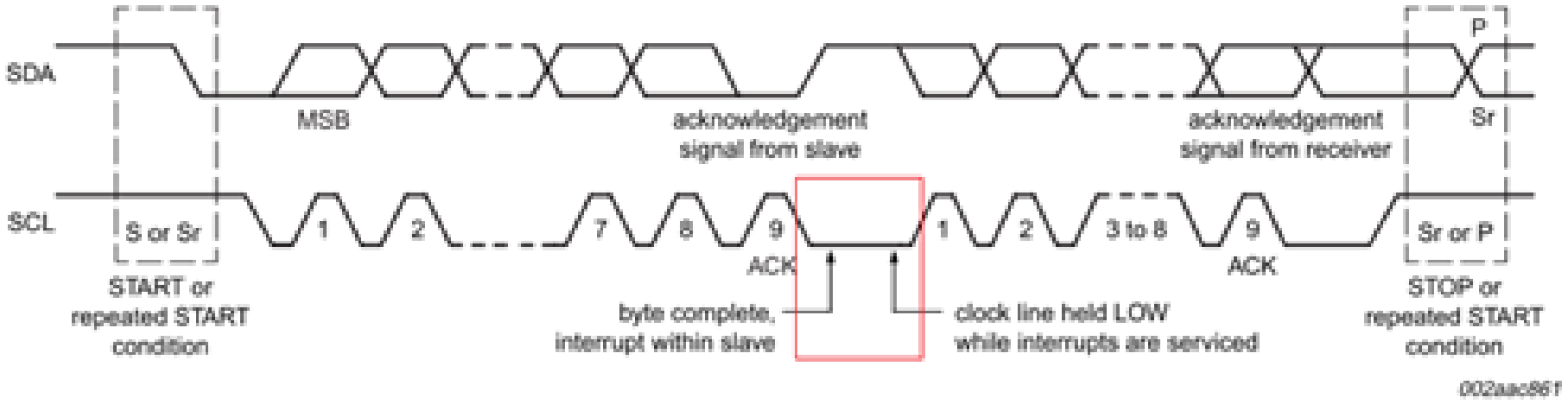
# I2C Communication

- Master begins the communication by Issuing the Start condition (S).
- Master sends unique 7-bit slave address with (MSB) first. The 8th bit which is the $R/\overline{W}$ bit specifies whether the slave is now to receive or transmit.
- ACK bit is issued by receiver for previous byte.
- Transmitter (Slave or Master) transmits a byte of data starting with MSB.
- Receiver (Slave or Master) issues ACK at the end of the byte.
- Pattern is repeated if more bytes need to be transmitted.
- In a Write transaction (Slave Receiving), when the Master is done transmitting all the bytes it wants to send, it monitors the last ACK and then issues the last STOP condition.
- In a Read transaction (Slave Transmitting), the Master does not acknowledge the final byte it receives. This tells the slave that its transmission is done. The Master then issues the STOP condition.



# I2C Clock Stretching

- I2C devices can slow down communication by stretching SCL.
- This is achieved by the I2C device by holding down SCL to prevent it from rising high again.
- This allows us to slow down the SCL clock rate or stop I2C communication for a while.




Source: https://e2e.ti.com/support/microcontrollers/msp430/f/166/t/120415

# Advantages/Disadvantages of I2C

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- Advantages:
  - Only uses two wires.
  - Supports multiple masters and multiple slaves.
  - ACK/NACK bit gives confirmation that each frame is transferred successfully.
  - Hardware is less complicated than with UARTs.
  - Well known and widely used protocol
- Disadvantages:
  - Slower data transfer rate than SPI.
  - The size of the data frame is limited to 8 bits.
  - More complicated hardware needed to implement than SPI.

# Audio

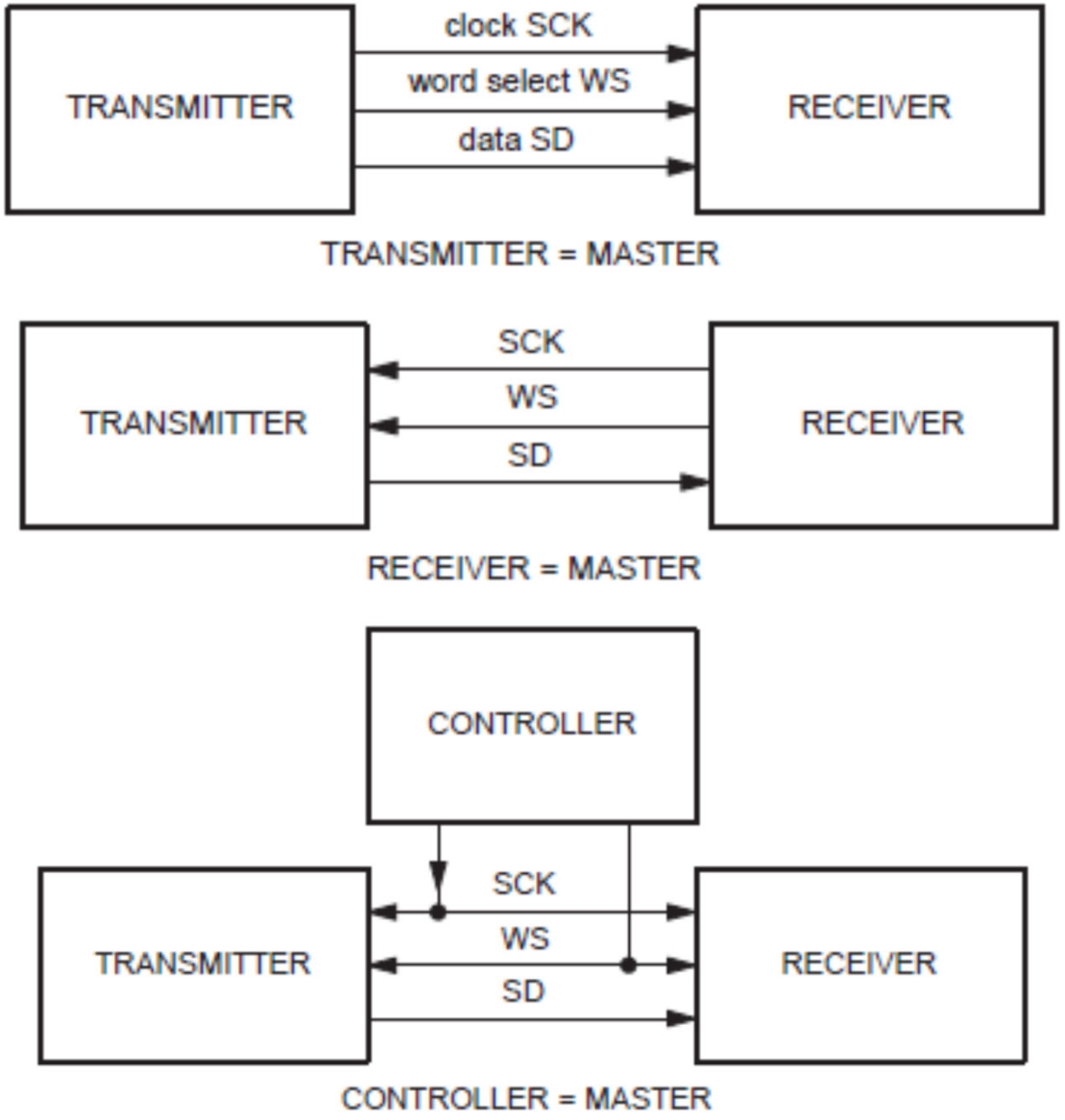
# I2S

- Many digital systems have been introduced to the consumer audio market like compact disc, digital audio tape, digital sound processors, digital TV-sound, etc.
- The digital audio signals in these systems are being processed by a number of IC chips such as:
  - A/D and D/A converters
  - DSPs
  - Error Correction Chips
  - Digital Filters
  - Digital I/O Interfaces
- Standardized communication structures are vital for both the equipment and the IC manufacturer to increase system flexibility.
- I2S (Inter-IC Sound) – A serial link especially for digital audio

# I2S – Bus Requirements

- The system only has to handle audio data.
- It consist of 3 lines
  - SCK – Clock
  - WS – Word Select
  - SD – Data
- The transmitter and receiver have the same clock signal for data transmission.
- The transmitter as the Master has to generate the bit clock, word-select signal and data.
- In a complex system, there may be several transmitters and receivers, which makes it difficult to define the Master.
- In such systems, there is usually a System Master controlling digital audio-flow between the various IC's.

# I2S – System Configurations



# I2S – Serial Data

- Serial data is transmitted in 2's complement with the MSB first.
- The MSB is transmitted first because the transmitter and receiver may have different word lengths.
- It isn't necessary for the transmitter to know how many bits the receiver can handle, nor does the receiver need to know how many bits are being transmitted.

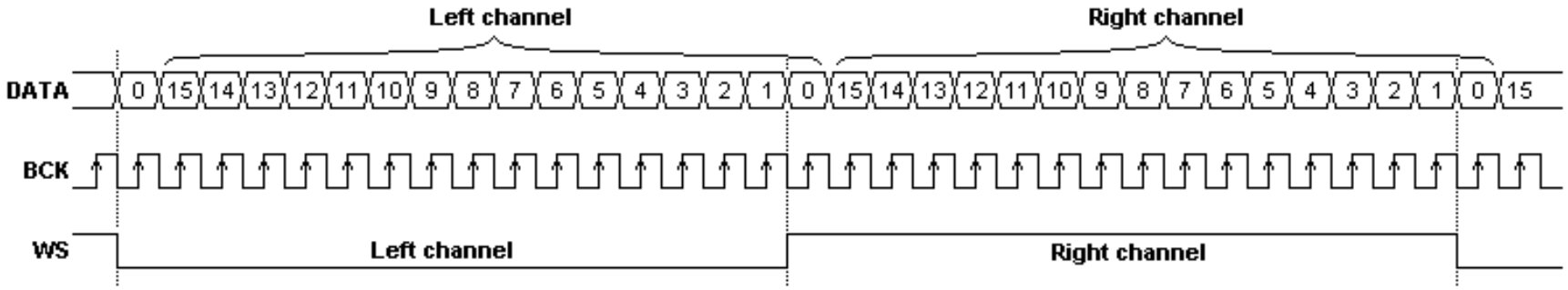
So how does the protocol handle different data packet sizes?

# I2S – Serial Data

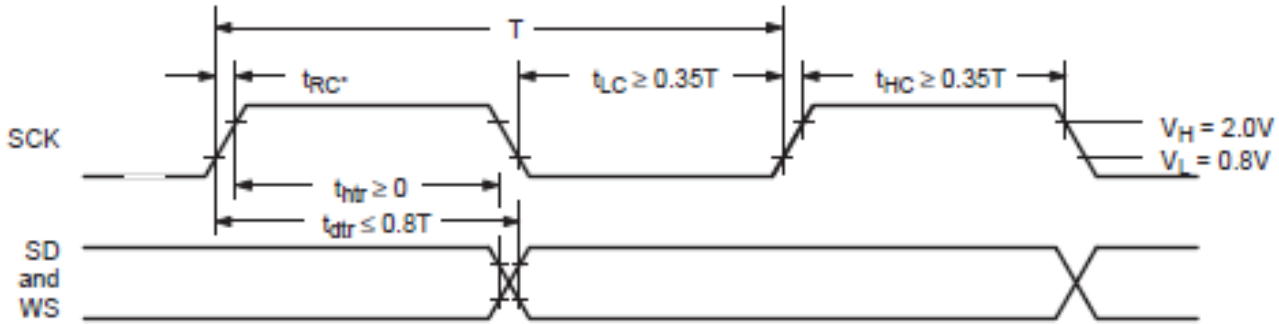
- If the receiver is sent more bits than its word length, the bits after the LSB are ignored.
- If the receiver is sent fewer bits than its word length, the missing bits are set to zero internally.
- So, the MSB has a fixed position, whereas the position of the LSB depends on the word length.

# I2S – Serial Data

- The transmitter always sends the MSB of the next word one clock period after the WS changes.




- As always, timing parameters are critical and must be obeyed.



# Infrared (IR) Communication

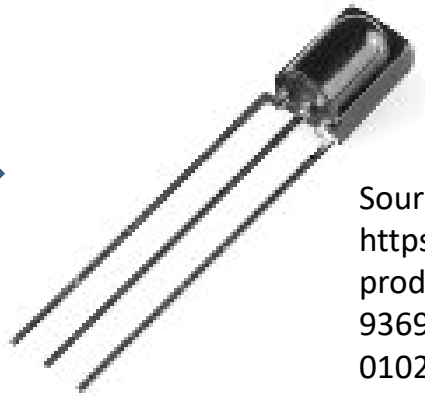
# Infrared (IR) Communication

- IR communication is a common, inexpensive, and easy to use wireless communication technology.
- IR light is very similar to visible light, except that it has a slightly longer wavelength. This means IR is undetectable to the human eye - perfect for wireless communication.





IR LED (Transmitter)

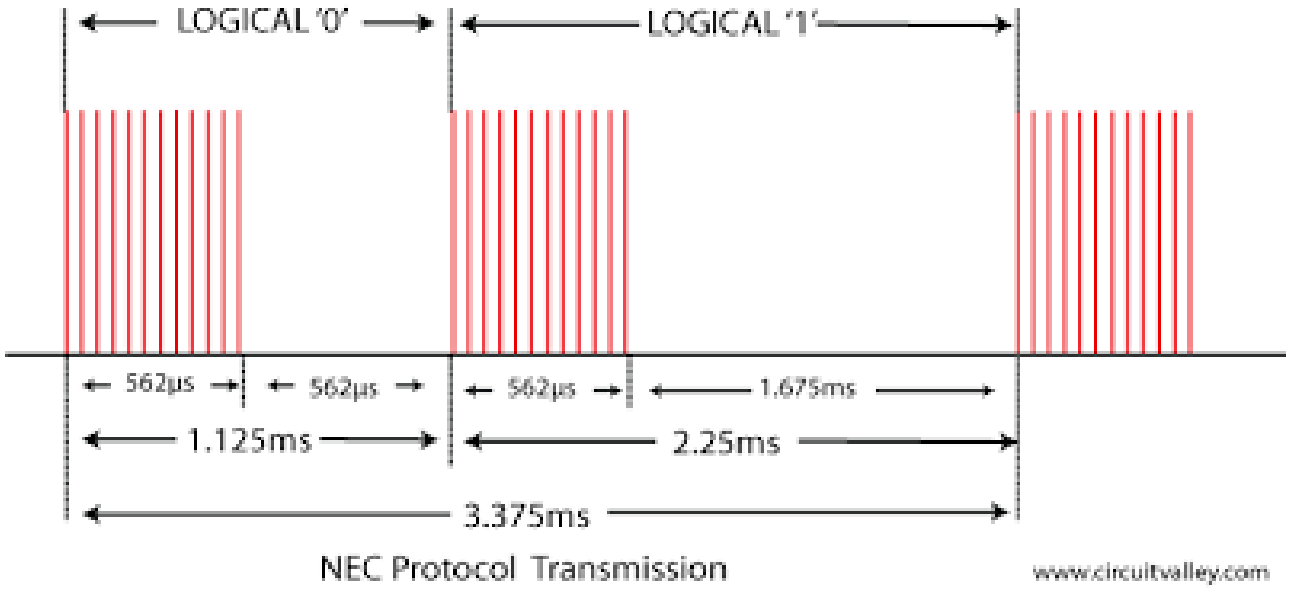


IR Receiver

Source: https://www.sparkfun.com/products/10266?_ga=2.173936995.1538986961.1497701026-1651978710.1495942862

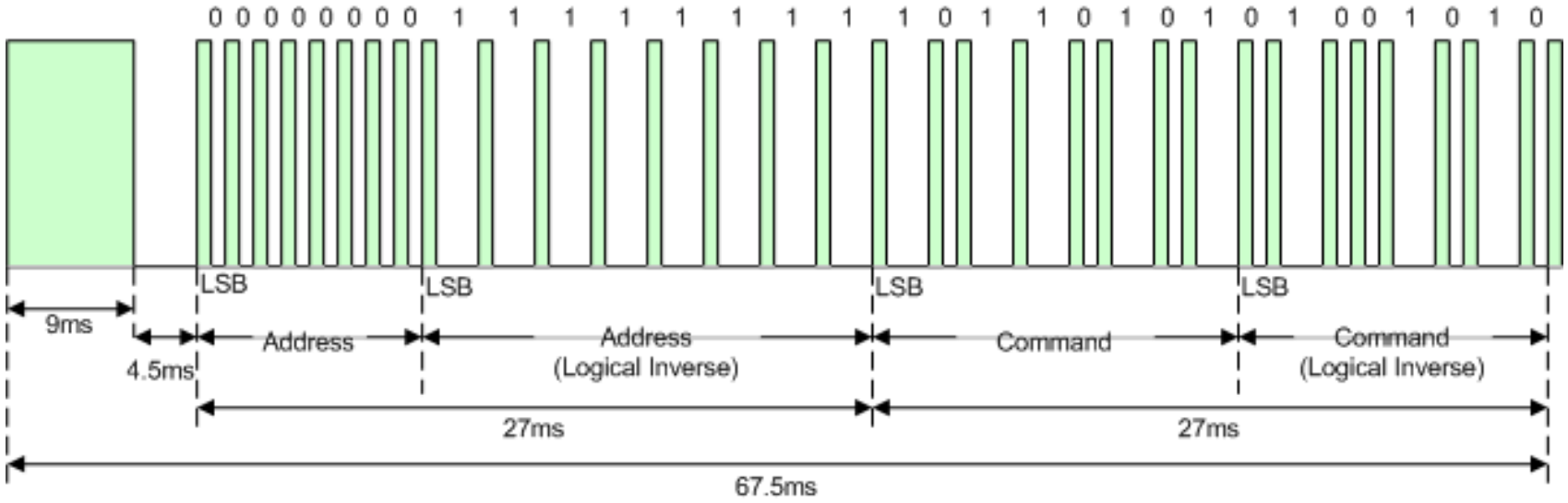
# NEC IR Transmission Protocol

- The NEC IR transmission protocol uses **pulse distance encoding** of the message bits.
  - Logical '0' – a 562.5us pulse burst followed by a 562.5us space, with a total transmit time of 1.125ms
  - Logical '1' – a 562.5us pulse burst followed by a 1.6875ms space, with a total transmit time of 2.25ms



# NEC IR Transmission Protocol (Cont.)

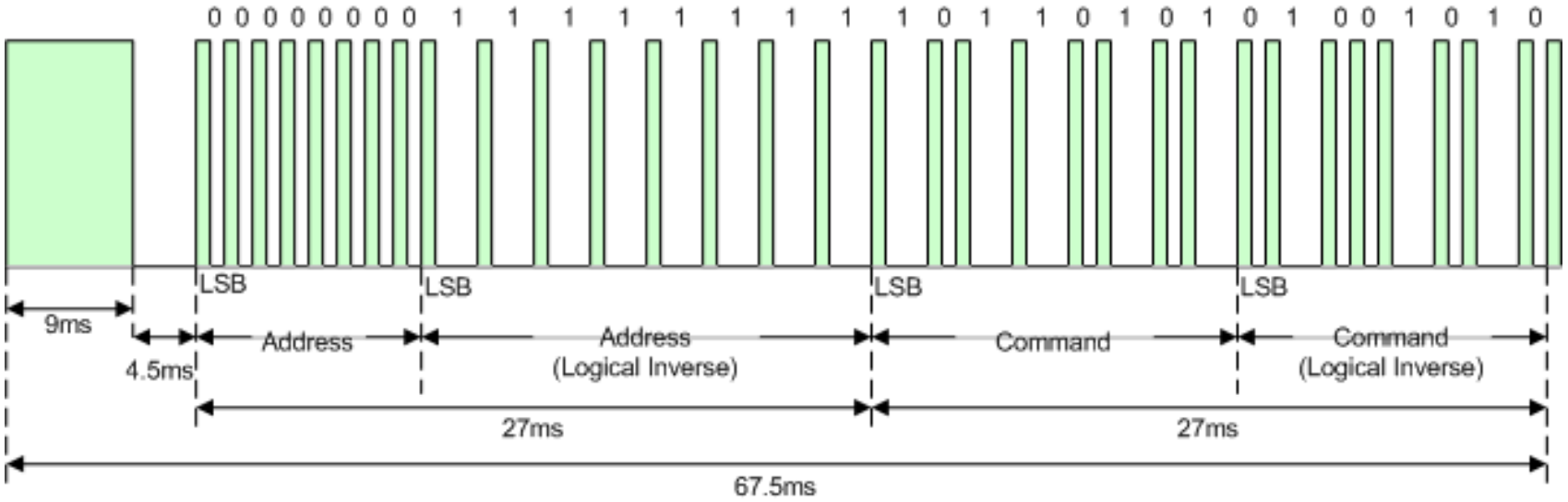
- When a Key is pressed on the remote controller, the data is transmitted as follows:
  - A 9ms leading pulse burst
  - A 4.5ms space
  - The 8-bit address for the receiving device
  - The 8-bit logical inverse of the address
  - The 8-bit command
  - The 8-bit logical inverse of the command.
  - A final 562.5us pulse burst to signify the end of the message transmission.




Source: http://techdocs.altium.com/display/FPGA/NEC+Infrared+Transmission+Protocol

# NEC IR Transmission Protocol (Cont.)

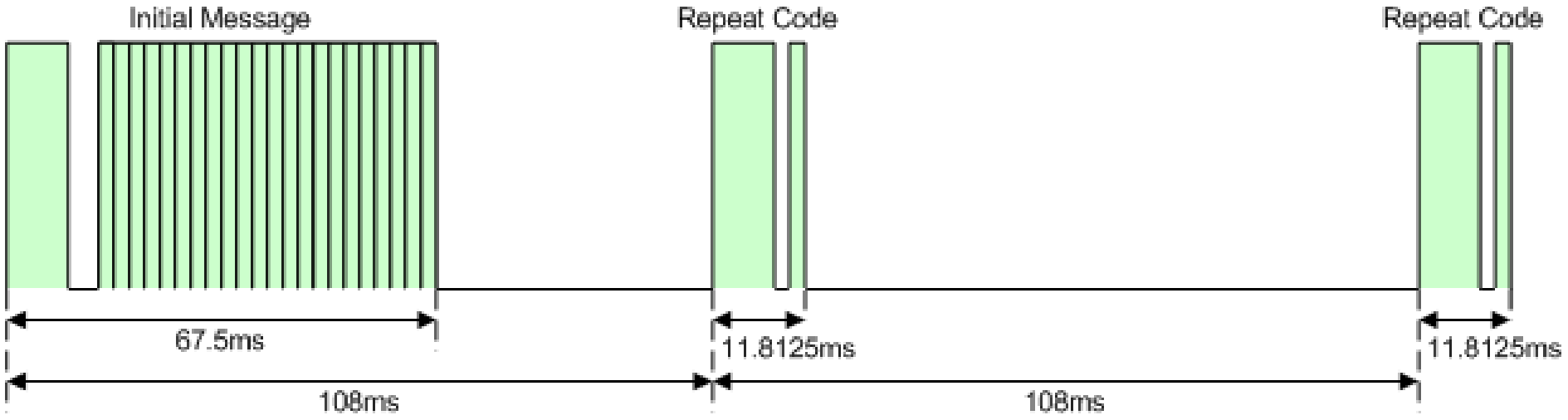
- The four bytes of data bits are transmitted LSB first.
- The figure below illustrates the format of an NEC IR transmission frame, for an address 0x00 ($0000\ 0000_2$) and command of 0xAD ($1010\ 1101_2$).
- Address and Command are transmitted twice. The second time all bits are inverted and can be used for verification of the received message.




- It takes 67.5ms to fully transmit the message frame (excluding the final 52.5us pulse burst that signifies end of message).

# Repeat Codes

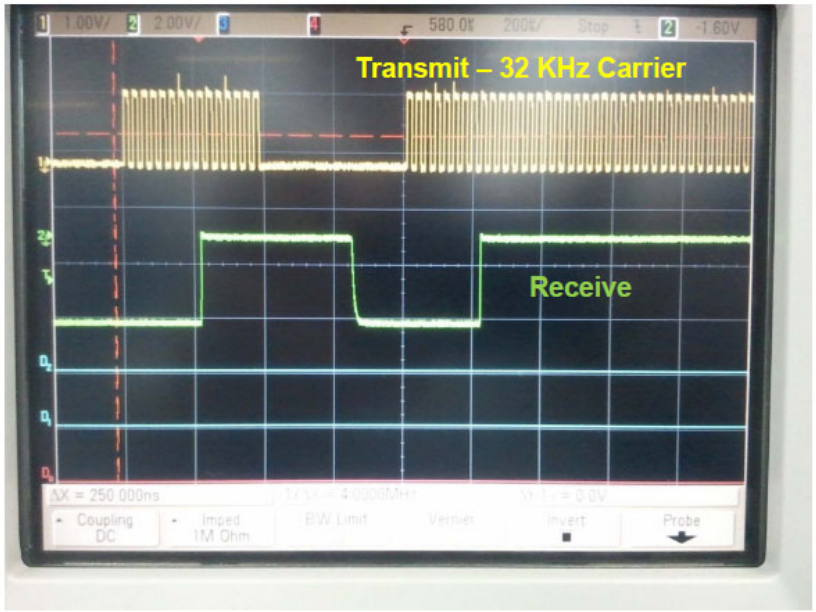
- If the key on the remote controller is kept depressed, a repeat code will be issued, typically around 40ms after the pulse burst that signified the end of the message.
- The repeat code will continue to be sent out at 108ms intervals, until the key is finally released. The repeat code consists of
  - A 9ms leading pulse burst
  - A 2.25ms space
  - A 562.5us pulse burst to mark the end of the space




Source: http://techdocs.altium.com/display/FPGA/NEC+Infrared+Transmission+Protocol

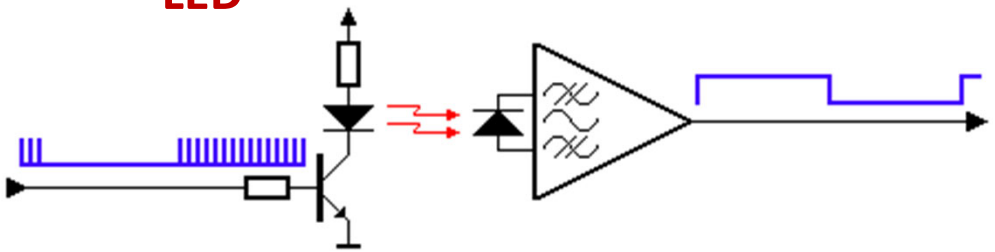
# Carrier Modulation

- Transmitted IR signal needs to be modulated so that the IR receiver is able to identify the signal among all the ambient IR.
- Modulating a signal is like assigning a pattern to your data, so that the receiver can pick up the transmitted data. This results in higher transmission fidelity.
- A common modulation scheme for IR communication is 38KHz modulation.




**Modulated signal driving IR Transmitter LED**

**Demodulated signal at receiver**

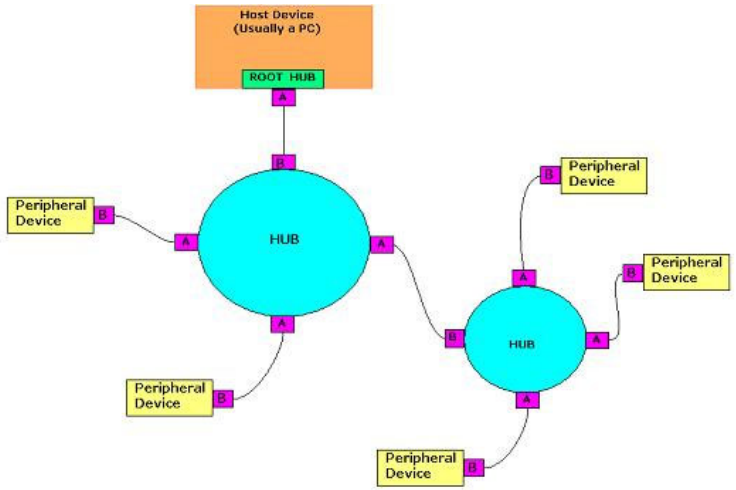


Source: http://www.sbprojects.com/knowledge/ir/

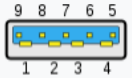
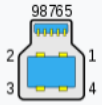
# Universal Synchronous Bus

# USB

- Universal Synchronous Bus
  - Set of interface specifications for high speed wired communication between electronics systems peripherals and devices with or without a PC/computer.
  - Major goal was to define an external expansion bus to add peripherals to a PC in an easy and simple manner.
  - Supports Hot-Swapping where devices can be plugged and unplugged without rebooting the computer or turning off the device.



# USB Versions

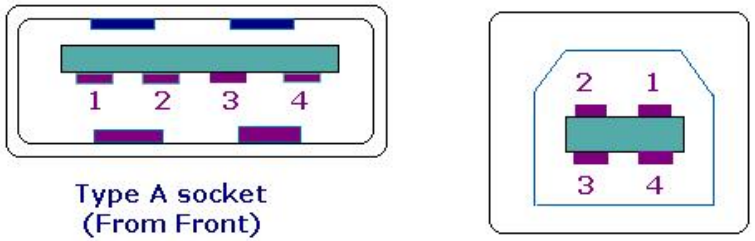
| Connectors | USB 1.0 1996 | USB 2.0 2001 | USB 2.0 Revised | USB 3.0 2011 | USB 3.1 2014 | USB 3.2 2017 | USB4 2019 |
|---|---|---|---|---|---|---|---|
| Data rate | 1.5 Mbit/s (*Low Speed*) | 480 Mbit/s (*High Speed*) | | 5 Gbit/s (*SuperSpeed*) | 10 Gbit/s (*SuperSpeed+*) | 20 Gbit/s (*SuperSpeed+*) | 40 Gbit/s (*SuperSpeed+ and Thunderbolt 3*) |
| | 12 Mbit/s (*Full Speed*) | | | | | | |
| Standard | Type A   Type-A | | | Type A   Type-A SuperSpeed | | Deprecated | |
| | Type B   Type-B | | | Type B   Type-B SuperSpeed | | | |
| | N/A | Type C (*enlarged*)  | | | | | |

Mini, micro

Source: https://en.wikipedia.org/wiki/USB

# USB – Connectors and Power

- The USB standard uses cables with two distinct connectors.
  - Type A – These connectors head upstream toward the computer
  - Type B – These connectors head downstream and connect to individual devices



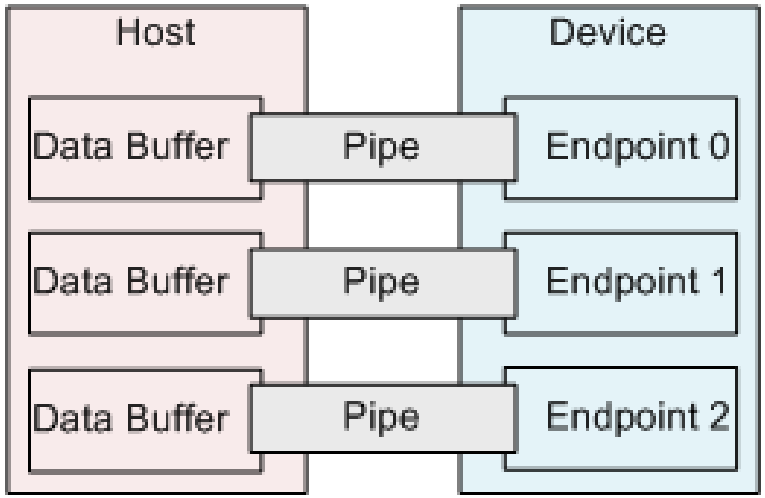

USB SOCKETS & PINS

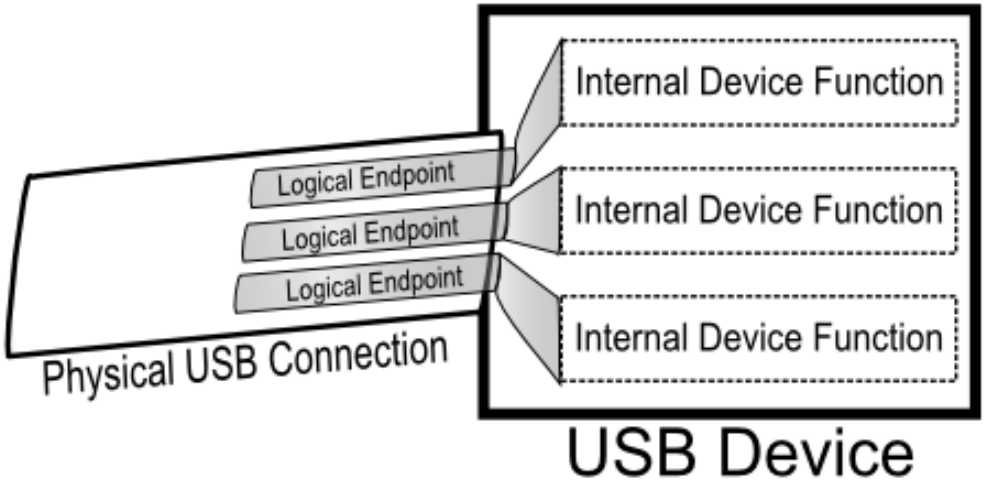
- Inside the cable, there are two wires to supply power and a twisted pair for the data.
- On the power wires, the computer can supply up to 500mA of power at 5V.
- If a peripheral device needs more than that, then it needs its own power supply.

| Pin No. | Signal | Color |
|---|---|---|
| 1 | + 5V Power | Red |
| 2 | - Data | White/Yellow |
| 3 | + Data | Green / Blue |
| 4 | Ground | Black / Brown |

# USB – Communication

- When a USB peripheral device is first attached to the network, a process called enumeration gets started.
- During enumeration, the device information is acquired and it is assigned a unique 7-bit address.
- Communication actually takes place between the host and endpoints located in the peripherals.
- An endpoint is a uniquely addressable portion of the peripheral that is the source or receiver of data.
- Four bits define the devices endpoint address.



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1 August 2022